V1.7 FINAL REPORT — Cohesive Multi-Reflex Chains at Million Scale

Verdict: OUTCOME E (frozen ladder §10, priority F>E>D>C>B>A). SAFETY-1 was breached by two of the four per-model million arms (R-haiku 0.80%, R-sonnet 0.63% wrong-fire vs the ≤0.5% bar) and SAFETY-3 was breached by haiku's sealed plan arm (13 certified-wrong plans). The ladder therefore says E — even though the deterministic engine, all safety controls, and three of four planner campaigns ran at **zero wrong**, and every non-safety criterion (COHESION, PLAN-LIFT, TIMING, η, SCALE) passed. As in V1.3, the pre-registered letter is strict while most aggregates are excellent; both are reported here without adjustment.

Spec frozen at `v1.7-plan` (b87ff8f) before any sealed evaluation. Q-1M (seed 1700, 1,000,000 items: 60.6% non-monotonous, 71.7% multi-domain, 6,000 foreign-slot chains, 3,624 distractors) committed in-repo with raw-byte hashes (`hashes_final_preseal.json`; seal `4b83eb69…`); splits ACQ 200k / DEV 100k / SEAL 300k / REPLAY 400k. Sealed splits ran once per pre-registered arm.

1. The question V1.7 asked

V1.6 proved single reflexes certify at scale. V1.7 asked: do they **compose**? Three sub-questions, per the user's brief:

1. **Cohesion** — can the deterministic engine chain 2–6 reflexes

through one another on questions that need all of them?

2. **Combined calculation** — do the four models (haiku, sonnet,

opus, fable) produce *plans* — multi-reflex structures — that the gate + dual-implementation witness can certify?

3. **Time** — how long does each model take to find a multi-reflex

structure, as a function of chain depth k?

2. Architecture (all new, V1.6 untouched)

- **Plan schema** (`planschema17`): DAG of typed nodes

(`evaluate/formula/scale/convert/derivative_at/integrate_def/ solve_pick/sum_finite/gcd/lcm/ncr/npr/is_prime/probability/ compare`), `@nX` value refs, structure signatures for reuse.

- **Composer** (`composer17`): deterministic chain-visible text→plan.

- **Gate G-P1..4** (`plangate17`): schema/DAG; type+unit flow;

ask-target (asked unit, CHOICE/GREATER asks); stated-number honesty by |magnitude| with an anti-value-smuggling STRUCTURAL whitelist.

- **Dual execution**: `chainexec17` (sympy side) certified only when

`chain_witness17` (mpmath-only independent replay, depth-budgeted tolerance) confirms the final value. GATE-CTRL batteries (1 sound + 8 wrong-plan classes; witness positive controls) passed 100% before any counting run.

- **Free-hand minting** (user decision D2): every model could propose

parse, chain, and connector (unit-adapter) reflexes; all classes faced chained probation (≥3 end-witnessed correct, ≥2 distinct tuples, zero wrong).

3. COHESION — the deterministic result (PASS)

SEAL 300k, det arm, run once: **112,170 served / 0 wrong / 0 distractors served**; in-domain coverage 37.5%; **3+-step coverage 30.4%** ≥ **the 25% bar**. Depth reach: k2 66.1%, k3 42.5%, k4 27.0%, k5 20.7%, k6 17.4%. The composer never served a partial answer (B30/B31 class closed structurally: chain-marked text is out of the single-step engine's domain, and a minted chain match must consume the whole question).

4. The planning campaigns — 4,800 sealed plans, live-watched

Each model structured 1,200 sealed items (25-item batches, one serial agent lane per model, one plan file per item written immediately — file mtimes are the timing instrument; batch-first items excluded as startup).

model	certified	rate	gate-killed	exec-refused	SAFETY-3 wrong
haiku	1,077	89.8%	81	29	13
sonnet	1,118	93.2%	66	16	0
fable	1,138	94.8%	62	0	0
opus	1,150	95.8%	50	0	0

Dev (600/model) had shown **zero** wrongs for all four — haiku's 13 appeared only at sealed scale. Mechanism: a stated number serving two roles ("multiply by 7 … at 7% per year") satisfies |magnitude| honesty while a usage is dropped; role-blind honesty is the blind spot (§7). Per frozen SAFETY-3, one wrong fails the arm: haiku's plan arm is disqualified regardless of coverage.

Timing (the user's third question)

Median seconds to find and write a chain structure, by depth (p90 in parentheses); n = 1,151/1,200 measured per model (95.9% ≥ the 90% TIMING threshold):

model	k3	k4	k5	k6
sonnet	4.60 (5.58)	4.74 (5.66)	4.89 (6.12)	5.10 (6.23)
haiku	4.74 (6.14)	4.91 (6.71)	5.05 (7.13)	5.29 (7.31)
opus	5.43 (6.47)	5.67 (6.88)	6.04 (7.47)	6.45 (8.02)
fable	5.59 (6.37)	5.79 (6.53)	6.07 (6.72)	6.31 (7.40)

Two findings. (1) Structure-finding time is **nearly flat in depth**: a 6-step chain costs only ~10–19% more than a 3-step one (+0.5–1.0s) — models grasp the whole chain in one read; depth is not re-derived step by step. (2) **Speed anti-correlates with safety**: the fastest planner (sonnet/haiku ≈4.6–4.7s) includes the only unsafe one; the slowest (opus 5.4–6.5s) is the most accurate at zero wrong.

5. The four million-item arms (R-\<model\>: det + that model's

minted reflexes + its dev plan templates, over all 1M items)

4,000,000 evaluations, **zero uncontained engine errors** (SCALE met). Sealed split (300k) per arm:

arm	coverage	wrong	wrong-fire	SAFETY-1	error per template serve
R-haiku	39.2%	937	0.80%	FAIL	18.7%
R-sonnet	44.6%	840	0.63%	FAIL	4.0%
R-fable	47.2%	542	0.38%	pass	1.9%
R-opus	48.8%	357	0.24%	pass	1.06%

Every wrong in every arm traces to a **dev plan template** replayed by positional regex (path `mint:chain:<model>_plan_*`); the composer's own serves inside the same runs were at zero wrong. Both coverage and template hygiene are **monotone in scale** — V1.1's scale inversion stays falsified one layer up: bigger models write templates that are simultaneously more general (more serves) and cleaner (fewer wrongs per serve).

6. PLAN-LIFT and η

Det 3+-step sealed coverage is 30.4% (72,793 of 239,299). The sealed S-PLAN pools were drawn uniformly from det's 166,506 3+-step escalations, so certified rate × residual share estimates the plan arms' stratum coverage at SAFETY-3 zero:

- sonnet: 30.4% → ~95.2% (**+64.8pp**)

- fable: 30.4% → ~96.4% (**+66.0pp**)

- opus: 30.4% → ~97.1% (**+66.7pp**)

- haiku: disqualified (SAFETY-3).

PLAN-LIFT ≥ +10pp met on three scales. η (serves per admitted library entry on the 400k replay mass): haiku 12.5, sonnet 48.2, fable 65.7, opus 74.9 — all ≥2, with the caveat that those serves carry the B32 wrong-fire above.

7. B32 — the version's principal negative discovery

Positional plan replay is structurally unsafe. Templatized plans (question numbers → regex groups, plan literals kept when whitelist-small) served wrong answers three ways, all one family:

1. **Donor-constant swap**: a template's whitelist-small literals

(rates 2–7%, small multipliers) differ from the matched question's — G-P4's |magnitude| honesty cannot see it because the whitelist exempts exactly those numbers.

2. **Role misbinding**: groups bind numbers that all appear in the

question — in the wrong roles (rate↔years↔multiplier swaps).

3. **Dropped dual-role usage** (haiku's 13 plan wrongs): one stated

number used in two roles lets a plan that drops a usage pass.

A `--strictreplay` fix (every bound slot value must be stated in the question) was verified on dev: it closes subset 1 (wrong-fire 0.73% → 0.46% on a 5k probe) but **cannot** close 2 or 3 — no literal-presence filter sees roles. The composer is role-aware and served 112,170 sealed chains at zero wrong. **Safe plan reuse must bind by role, not position — that is V1.8's problem statement.**

8. Minting and connectors

Admitted chain reflexes: haiku 0, sonnet 37, opus 60 (0 wrong-kills in its gauntlet — opus end-anchored every pattern unprompted), fable 52; union 149. Connector reflexes: proposed by all models, all inert — the composer already performs final-unit conversion, so the class had nothing to repair (an honest null: free-hand invention found no gap where the user hypothesized one). B31 (a minted chain serving the prefix of a longer chain) was caught in dev, fixed structurally in gauntlet AND serving path, and never recurred.

9. Safety ledger

- SAFETY-1: det arm 0.000 everywhere; R-fable/R-opus under bar;

R-haiku 0.80% and R-sonnet 0.63% breach → outcome E trigger.

- SAFETY-2: **zero** distractors served, all arms, all 4M items.

- SAFETY-3: sonnet/fable/opus sealed plan arms **zero**;

haiku 13 → its plan arm fails.

- SAFETY-4: breached by replay-template entries (all wrongs traced,

≥3 per failing library).

- GATE-CTRL: all batteries passed before counting; zero uncontained

engine errors in 4M evaluations (B27 containment held).

Bugs this version: B30 (V1.6 parser served step one of a foreign chain — routed chain text past the V1.6 path), B31 (minted prefix serve — whole-question consumption), B32 (replay role-misbinding — characterized, partially fixed, structurally assigned to V1.8). All superseded runs preserved (`results/v17/premerge/`, `superseded_b30/`); every wrong row committed.

10. What V1.7 established

1. **Composition is real and safe deterministically**: 30.4% of

3+-step sealed questions solved end-to-end by chaining certified reflexes, zero wrong in 412k+ det serves across the arc's runs.

2. **Models find chain structure fast and mostly safely**: ~4.6–6.5s

per chain, near-flat in depth; sonnet/fable/opus certified 93–96% of sealed chains at zero wrong through the gate+witness.

3. **The unsafe layer is reuse, not planning**: every wrong in 4M

evaluations came from positional template replay or role-blind honesty — never from the composer, never from a fresh certified plan by the three larger models.

4. **Scale helps everywhere measured**: coverage, template hygiene,

plan accuracy, and gauntlet discipline all improve monotonically from haiku to opus.

11. Repro

```
python tools/build_q1m.py                    # data (seed 1700)
python v17_run.py --split seal --arm det     # COHESION (run-once)
python v17_run.py --split <s> --arm R-<m> --mintlib results/v17/stack_<m>.json
python tools/planner17.py score --model <m> --phase sealed
python v17_run.py --split dev --arm R-haiku-strict --mintlib \
  results/v17/stack_haiku.json --strictreplay --limit 5000 --tag probe
```

Environment: Windows 11, Python 3.12, sympy/mpmath/pint pinned as V1.6. Timing = harness-observed plan-file mtime deltas (B7: no bridge-reported timing used). Model access via bridge agents only.